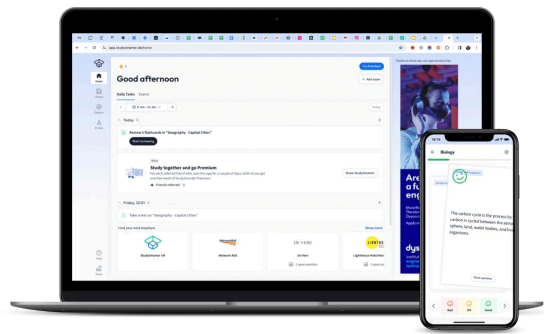


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Sampling Theorem

Dive into the fascinating world of computer science and explore the foundation of how data is represented digitally through an immersive understanding of the Sampling Theorem. Your journey will start with grasping the basic principles of this theorem before examining its intimate relationship with data representation. You'll delve deeper into the subtleties and complexities of the Nyquist Shannon Sampling Theorem and learn how to determine the Nyquist Theorem Sampling Rate accurately. Further investigation will lead you to the explorative study of the Sampling Theorem's formula and techniques. Ultimately, practical applications of these principles are exposed using hands-on Sampling Theorem examples. This in-depth look into the Sampling Theorem will pave the way to a comprehensive mastery of [data representation in computer science](#).



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What is the core principle of the Sampling Theorem in Computer Science?

The Sampling Theorem, Also Known As The Nyquist-Shannon Theorem, Provides A Fundamental Rule For Converting Continuous-Time Signals Into Discrete-Time Signals. It States That A Signal Can Be Perfectly Reconstructed If The Sampling Frequency Is More Than Twice The Highest Frequency Component Of The Signal.

The Sampling Theorem Is Only Applicable In The Conversion Of Analog Audio Signals From Analog To Digital Format.

The Principle Of Sampling Theorem Involves The Conversion Of Binary Data Into Real-World, Continuous Signals.

The Sampling Theorem States That Any Signal Can Be Accurately Digitised By Keeping The Sampling Frequency Lower Than Twice The Highest Frequency Component Of The Signal.

What is Quantisation and how does it relate to the Sampling Theorem?

Quantisation Is The Process Of Approximating Continuous Values To Meet The Requirement Of The Sampling Theorem.

Quantisation Is The Process Of Mapping Continuous Values To A Large Set To Output Values In A Sampled Format. It Involves A Rounding-Off Process. It Is Vital In Implementing The Sampling Theorem, But It Introduces Quantisation Error In Signal Representation.

Quantisation, Relevant In Signal Processing, Involves Approximating Any Errors Generated During Sampling To A Finite Set Of Values.

Quantisation Is Dividing The Sampled Values Into Discrete Steps To Ensure Perfect Reconstruction Of The Original Signal.

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Understanding the Sampling Theorem in Computer Science

The Sampling Theorem, also known as the Nyquist-Shannon theorem, provides the fundamental bridge between continuous-time signals (analog) and discrete-time signals (digital). It's an essential concept in the realm of Computer Science, especially when dealing with [signal processing](#), [data compression](#), and multimedia applications.

You may think of it as the rule book for converting real-world, continuous signals into a format that computers can understand and process.

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Definition

Digging into the details, the Sampling Theorem states that a signal can be perfectly reconstructed from its samples if the sampling frequency is more than twice the highest frequency component of the signal.

Let's break this down a [bit](#):

- Signal: Anything that carries information, like sound waves, light waves, or radio waves.
- Frequency: It is the rate at which something occurs. It is measured in Hertz (Hz).
- Sampling frequency: The number of samples taken per second. Also known as the [sample rate](#).

Below is the formal representation of the Sampling Theorem:

$$f_s > 2f_m$$

Example

In which f_s stands for the sampling frequency and f_m for the maximum frequency of the signal.

Deep dive

Considering sound as an example, the human hearing range is approximately 20 Hz to 20,000 Hz. Hence, according to the Sampling Theorem, to digitally reproduce sound that covers the whole range of human hearing, you must sample at least at a frequency of 40,000 Hz.

Relation between Sampling Theorem and Data Representation

The Sampling Theorem underpins the digitisation of signals that has made digital storage, processing, and transmission possible - core aspects of modern computing.

Definition

Since computers operate on binary data (0's and 1's), the Sampling Theorem allows us to convert real-world, continuous signals into discrete binary data that a computer can understand.

A fundamental process related to the Sampling Theorem in Computer Science is [Quantisation](#). It's the process of mapping input values from a large set (often a continuous set) to output values in a (countable) smaller set. Below is a simple example of the Quantisation process:

Original Signal: 12.8, 15.2, 18.1, 14.9
Quantised Signal: 13, 15, 18, 15

You can see [Quantisation](#) as a rounding-off process. It's vital in digitising signals but introduces a [Quantisation](#) error in the signal representation.

Example

For example, an audio file in a raw, uncompressed form can be huge. By using the Sampling Theorem and further applying Quantisation and Coding, we can significantly compress the size of the audio file, making it easier to store and transmit.

Delving into the Nyquist Shannon Sampling Theorem

The Nyquist Shannon Sampling Theorem is recognised universally as the guiding principle for digitally capturing continuous signals. It is named after Harry Nyquist and Claude Shannon, two prominent figures in the field of information technology and telecommunications.

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How Nyquist Shannon Sampling Theorem Enhances Data Representation

When it comes to understanding how the Nyquist Shannon Sampling Theorem enhances data representation, we need to dive deep into the realms of [signal processing](#) and [data encoding](#). In essence, this theorem charts a path for transforming continuous signals into digital, without any loss of information. This happens as long as there's adherence to a vital parameter: the **sampling rate**.

The term **sampling rate**, also known as sample frequency, denotes the number of times a signal is measured or "sampled" per second. To replicate a signal without loss, this theorem suggests that the sampling frequency should be more than twice the signal's highest frequency. This criterion is often labelled as the **Nyquist rate**, and one must make sure that the signal doesn't contain frequency components higher than this rate. If such components exist, a phenomenon called **aliasing** occurs, leading to distortions.

Example

For instance, in the case of audio signals imperceptible to humans above 20kHz, the theorem suggests that digital audio should be sampled at least at 40kHz to accurately reproduce the sounds.

Signals captured according to the theorem turn into binary numbers, allowing an accurate digital representation. This digital representation opens up possibilities of effective alignment, sorting, classification, and [compression](#).

Piecing Apart the Components of Nyquist Shannon Sampling Theorem

The Nyquist Shannon Sampling Theorem is fundamentally built on two concepts: **sampling** and **aliasing**. To implement the theorem effectively, one must understand these components.

Sampling refers to the process of capturing a signal's value at uniform intervals to create a sequence of samples. Each sample represents the value of the signal at that specific instance. These samples are then coded into binary format and used as a base for various digital applications.

In the process of digitization, the **aliasing effect** is a distortion that appears when higher frequencies in the original signal start to mimic lower frequencies after sampling. This effect occurs if one doesn't strictly maintain the Nyquist rate.

Sr. No.	Part of Nyquist Shannon Sampling Theorem	Description
1	Sampling	The conversion of continuous signal into discrete form by capturing the signal value at uniform intervals
2	Aliasing	An effect that can distort sampled signals when higher frequencies are incorrectly interpreted as lower frequencies

The Sampling Theorem brings its strength to the digital realm. It allows lossless conversion of continuous real-world signals into a form that computers, digital media players, computer networks, and other digital systems can work with.

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Detailed Analysis of the Proof of Sampling Theorem

The proof of the Sampling Theorem, or the Nyquist-Shannon Theorem, empowers us with a deeper comprehension of this profound aspect of Computer Science. It elucidates how we can recover an original signal from its samples, provided the sampling was done appropriately. To truly decipher its implications, let's break down the proof and its importance.

Breaking Down the Proof of Sampling Theorem

Bulk of the proof is algebra-based. It utilises the basics of Fourier Transform and Euler's formula to establish the theorem. The theorem articulated with f_s as the sampling frequency and f_m being the maximum frequency of the signal is as follows:

$$f_s > 2f_m$$

For the proof, we take a signal $x(t)$ which is band-limited to f_m , indicating it has no frequency components above f_m . The equation for $x(t)$ can be represented as the inverse Fourier transform of $X(f)$, its frequency spectrum.

substitutes t with nT in the equation above. This results in:

$$x(nT) = \int_{-f_m}^{f_m} X(f)e^{j2\pi f nT} df$$

The equation can be further simplified using the principles of algebra and Euler's formula.

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Importance and Implications of the Proof in Sampling Theorem

The proof of the Sampling Theorem is more than a mathematical conquest. It forms the theoretical underpinning for the digitisation of signals, a cornerstone in modern computing, digital communication and multimedia processing. The principal takeaway from the proof is the Nyquist criterion of having the sampling frequency f_s more than twice the maximum frequency f_m of the original signal. This understanding is ever-present in the design of digital systems, mainly when transforming analog to digital signals.

- Data Compression:** As elaborate sampling can lead to extensive data, understanding how to sample effectively paves the way for valuable data [compression](#) techniques.
- Anti-aliasing Filters:** Before sampling a signal, engineers often use filters to eliminate frequencies above f_m . This prevents aliasing, a pervasive issue during signal digitisation.
- Telecommunication & Broadcasting:** The exact replication of signals is crucial here. Sampling Theorem serves as the fundamental guideline, ensuring information conveyed isn't lost or distorted.
- Medical Imaging:** Devices like MRI scanners leverage the theorem to capture signals from the human body and reconstruct them digitally for analysis.

In summary, the proof of the Sampling Theorem doesn't just illuminate the theorem's tenants but equips you, the Computer Science scholars, with a crucial tool in the digital toolbox. Understanding this helps us appreciate the harmony between mathematics and technology in creating the digital world as we know it today.

Determining the Nyquist Theorem Sampling Rate

Fundamentally, the Nyquist Theorem offers us a precise yardstick to determine the sampling rate, a key parameter in signal conversion. It safeguards the integrity of the original signal and ensures a faithful digital representation. To identify the correct sampling rate, the theorem mandates that it should be at least twice the maximum frequency present in the signal.

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Understanding the Role of Nyquist Theorem Sampling Rate

The Nyquist-Shannon Theorem, or essentially the Sampling Theorem, constructs a bridge between the world of continuous time signals and its discrete counterpart. The heart of this theorem lies in the **sampling rate**, often termed as the sampling frequency.

Definition

The **sampling rate** is the frequency at which a signal is sampled per unit of time. It's often represented in Hertz (Hz).

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the sampling rate must be twice the maximum frequency of the signal.

The mathematical expression converging sampling frequency f_s and the maximum signal frequency f_m is:

$$f_s > 2f_m$$

Therefore, grasp the role of the Nyquist Theorem Sampling rate, as it's the guiding compass in maintaining fidelity during signal conversion. Remember that the sampling rate isn't a one-size-fits-all value - it must be maximised based on the characteristics and dynamics of each signal to ensure an accurate digital representation.

How Nyquist Theorem Sampling Rate Affects Data Representation

The role of Nyquist Theorem Sampling Rate becomes ever apparent when you delve into data representation. The ability to propose the 'right' sampling rate allows us to have precise, lossless data knowing that the digitised signal holds the essence of its continuous counterpart.

When we transform a real-world, continuous signal into a series of binary data, the samples collected act as a DNA blueprint of the signal, encapsulating its core information. These samples, coded into binary data, serve as a foundation for various digital applications.

Consider, for example, the act of recording sound. Each sound wave, which is a continuous signal, is sampled at regular intervals. These samples, or snapshots of the sound wave at some moment in time, are transformed into digital data which can be processed, stored, or even reproduced later.

It's critical to note that the quality of this digital sound will significantly depend on the chosen sampling rate. If you select a very high sampling rate, the binary representation will naturally be larger and more precise, but it might lead to wastage of storage space and unnecessary computational processing by containing more information than required. On the other hand, a low sampling rate might miss out on key frequency components, leading to lower quality playback or lossy data representation.

Thus, it's apparent that the Nyquist Theorem Sampling Rate and how it's determined influence the quality, size, and fidelity of digital data. It directs us in choosing the optimal balance between precision and resource consumption, playing a vital role in the efficient digital representation of continuous signals.

Exploration of the Sampling Theorem Formula

In the domain of Computer Science, the Sampling Theorem, or the Nyquist-Shannon Theorem, emerges as a cornerstone dictating the digitisation of signals. The entire theorem pivots around a mathematical formula that defines the premise of the theorem and sets guidelines for signal conversion in practice.

Getting to Grips with the Sampling Theorem Formula

When it comes to capturing real-world, continuous signals in the form of discrete data, one mathematical formula sets the ground rules - the quintessential formula of the Sampling Theorem. This theorem enunciates that a signal can be perfectly reconstructed from its samples if the sampling frequency is more than twice the highest frequency component of the signal. The formula can be represented as:

$$f_s > 2f_m$$

Here, f_s denotes the sampling frequency and f_m represents the maximum frequency component of the signal. This formula, albeit succinct, carries a profound implication. The Nyquist rate or $2f_m$ stands as the bare minimum sampling frequency required to ensure that the analog signal could be fully recovered from its samples. If the sampling rate is less than the Nyquist rate, it leads to aliasing, a phenomenon where different signals become indistinguishable from each other when sampled. However, one important point should not be overlooked. The formula assumes that the signal is band-limited, i.e., it doesn't carry frequency components above f_m . To better understand the elements of the formula, let us review them:

- **Sampling Frequency:** The number of samples obtained per second is referred to as the sampling frequency. It plays a pivotal role in capturing adequate information from the original signal. Higher the sampling frequency, greater will be the detail in which the original signal can be recreated.
- **Maximum Frequency:** It's the highest frequency component present in the signal. It's crucial to note that the quality and faithfulness of the reconstructed signal hinge on this factor as frequencies above this will not be recorded.

In essence, the Sampling Theorem Formula equips you with the ability to determine the minimum sampling frequency or rate that you should use to sample a particular signal without loss of information. Though it appears straightforward, it forms the backbone of the digital world, making multimedia applications, telecommunications, and even the internet a reality.

Significance of the Sampling Theorem Formula in Data Representation

As we delve into the realm of digital data representation, the contribution of the Sampling Theorem Formula is undeniable. From capturing images with your smartphones, watching digital TV, streaming audio files, or even playing video games, the propagation of this theorem is extensively noticeable. The formula essentially gives us the key to unlock the potential to convert real-world, complex, and continuous signals into a discrete set of data that computers can process. Representing data digitally has immense advantages, including the ability to process, store, reproduce, and transmit data efficiently. One of the remarkable aspects in computer science where the Sampling Theorem bloom is in Data Compression. Given that high-frequency sampling can yield substantial data, the theorem can guide us on optimal sampling. With the right balance on the sampling frequency, substantial data compression can be accomplished without losing out on crucial information.

Moreover, the theorem takes centre stage in shaping anti-aliasing filters. By designing filters to wipe off frequencies above f_m , we can prevent the effect of aliasing during signal sampling and ensure a faithful digital representation. In summary, the Sampling Theorem Formula is a key player in data representation in the digital realm. Understanding this formula and the theorem is vital in the field of computer science, and more broadly, for anyone dealing with digitisation of information. It's indeed transforming our world, one sample at a time.

Definition and Technique of Sampling Theorem

By now, it's becoming evident that the Sampling Theorem, or the Nyquist-Shannon Theorem, forms the core of digitisation. It's the bedrock that enables conversion of continuous, real-world signals into discrete data that digital systems can process.

Unpacking the Sampling Theorem Definition

Let's embark on a journey to decode the Sampling Theorem. It's a fundamental principle stating that a signal can be correctly decoded from its samples if the sampling frequency is greater than twice the maximum frequency of the original signal. This concept derives its roots from the extensive studies of signals and systems within the Computer Science realm. Analog signals are continuous by nature and can't be directly used by digital systems. However, continuous signals converted into discrete versions turn into data that computers understand. The conversion process involves evenly spaced sampling, which doesn't capture every single point of the analog signal in its entirety but enough to reconstruct the original without any loss of information. This transformation is possible only when the sampling frequency conforms to a specific parameter indicated by the Sampling Theorem. The formula stands as:

$$f_s > 2f_m$$

To recall, f_s denotes the sampling frequency, and f_m signifies the maximum frequency component in the signal. Bear in mind that the Sampling Theorem assumes that the signal is band-limited, pointing out it doesn't comprise frequency components above f_m . In essence, the definition of the Sampling Theorem zooms into the relationship between the sampling frequency selected and the highest frequency present in a continuous signal. It offers a pathway for maintaining signal fidelity during the conversion process and serves as an indispensable guide for digitising signals.

Learning the Sampling Theorem Technique and its applications

While the definition gives a glimpse of the theorem, the technique of the Sampling Theorem forms its practical backbone. Let's dissect the key aspects:

1. The first step involves determining the maximum frequency component f_m present in the signal.
2. Once the band-limitation of the signal is known, the sampling frequency or rate f_s is set in compliance with the theorem, which is more than twice the maximum signal frequency.
3. The signal is then sampled at this rate, resulting in a sequence of discrete data points or samples encapsulating the pertinent elements of the original signal.
4. These samples, encoded into binary, serve as the digitised form of the signal ready for use by digital systems.

The technique is not only tied to the Sampling Theorem but also illustrates its diverse applications. From sound recording, image processing, data compression, to fast-paced signal transmission over the internet, the theorem underpins the digital revolution. Here are some specific areas where the Sampling Theorem's technique is demonstrable:

- **Telecommunications:** In the modern world, most communication happens digitally. The theorem aids in transforming analog voice signals into digital data for transmission over networks, maintaining exactness and clarity of the information.
- **Audio and Video Encoding:** Whether it's about digital music or high-definition video, the theorem makes sure the media we consume retains high quality by guiding the selection of sampling rates when digitising these signals.
- **Image and Graphics:** Fundamental to digital imaging and graphics, the theorem allows capturing visual signals and transforming them into pixel data, contributing to modern digital photography and imaging technologies.
- **Data Compression:** Given the copious data generated through extensive sampling, the theorem provides insights into the optimal [sample rate](#) needed to represent the data efficiently without loss of crucial information, invaluable for data compression.

Now that you're acquainted with the definition and technique of the Sampling Theorem, the next step is to apply these learnings in practice. Understanding the Sampling Theorem and its process provides a solid foundation for comprehending essential concepts about digital data representation and paves your way to become effective problem solvers in the digital realm.

Practical Applications: Sampling Theorem Example

In consolidating what we've learnt about the Sampling Theorem, a real-world practical example stands as an astute teaching device. Let's dive into an example that brings the theory to life, demonstrating its distinct utility and impact.

Illustrating the Theory with a Sampling Theorem Example

To illuminate the principles of the Sampling Theorem, consider the task of digitally recording a piece of music or any audio signal for that matter.

This is where the Sampling Theorem comes into play. According to the theorem, to prevent any loss of information during the conversion process, the sampling frequency should be more than twice the highest frequency present in the sound signal.

For instance, the human ear can hear frequencies in the range of roughly 20 Hz (low-frequency sounds like a rumbling thunder) to 20,000 Hz (a very high pitch sound that many adults can't perceive). Thus, for a sound that spans the entire gamut of audible frequencies, the theorem suggests that the digital audio should be sampled more swiftly than twice the maximum audible frequency i.e., 40,000 Hz frequency. This is in line with the theorem's formula:

$$f_s > 2f_m$$

In actuality, most digital audio applications, such as CDs, sample audio at 44,100 Hz (well over the minimum 40,000 Hz stipulated by the theorem) for a [bit](#) of margin.

- The first step involves identifying the range of frequencies present in the sound.
- Next, apply the Sampling Theorem formula. For a safe margin, ensure that the sampling frequency is above 2 times the highest frequency component.
- Finally, apply this sampling rate while recording the audio, creating a faithful digital representation.

There you have it! A fine illustration reflecting the essence of the Sampling Theorem in actual application. This fundamental understanding acts as your guide, shaping the design and execution of all multimedia systems that engage digital audio processing.

Assessment of the Sampling Theorem Example's Impact in Data Representation

Deep dive

A simple audio recording task illustrates the stunning utility and profound impact of Sampling Theorem's principles on data representation.

The Sampling Theorem directs the way for faithfully capturing the characteristics of an analog audio signal in digital format. The digital representation not only allows recording but also enables easy transmission and storage of audio data over digital devices and networks.

The theorem's impact ebbs further as it contemplates how often we need to capture data. A sound signal, as in our example, is abundant with data in its raw, continuous form - picturing it as a sea of data won't be exaggerated. However, the sampling process necessitates collecting only significantly meaningful data at the rate proposed by the theorem. The resulting sampled data leads to a more efficient and structured representation, aiding smooth processing and comprehension by digital systems.

Example

A good illustration is the use of Sampling Theorem in CD audio. The audio, sampled at 44,100 Hz, saves key details while allowing for efficient data compression techniques that feed accurate, high-quality audio to our ears. In simple terms, without applying the Sampling Theorem, our music experience wouldn't have been the same!

Moreover, though not explicit in our example but vital to understanding, is that the theorem assists in preventing distortion or 'aliasing' that can occur when a signal containing high-frequency components, not catered to by an insufficient sample rate. By ensuring a sampling rate above twice the highest frequency, the theorem protects against loss of information, guaranteeing the truest representation of the original signal.

To conclude, the Sampling Theorem extends immense influence on data representation, emphasising its significance in computer science. It paves the way for efficient, comprehensive digital representation, shaping useful digital data from a sea of analog signals. Ultimately, it marks every step in our journey of experiencing the digital world - from the music you hear, the videos you stream, to the data you transmit.

$$f_s > 2f_m$$

Frequently Asked Questions about Sampling Theorem

- Q:** What is the significance of the Nyquist-Shannon Sampling Theorem in Computer Science?
- A:** The Nyquist-Shannon Sampling Theorem is significant in computer science as it provides a fundamental bridge between continuous-time (analog) signals and discrete-time (digital) signals. It underpins the design of digital communication systems and is critical for digital signal processing.
- Q:** What are the practical applications of the Sampling Theorem in Computer Science?
- A:** The Sampling Theorem underpins key practices in Computer Science, such as digital signal and image processing, data compression, numerical analysis, and artificial intelligence. It's crucial for converting analog signals into digital ones for further processing, a fundamental operation in computing.
- Q:** How does the Sampling Theorem contribute to the field of Digital Signal Processing in Computer Science?
- A:** The Sampling Theorem contributes to Digital Signal Processing by allowing analogue signals to be converted into digital signals without loss of information. It enables accurate reproduction of a continuous-time signal from its discrete samples, provided the sampling frequency is above a certain threshold.



Q: What is the fundamental principle behind the Sampling Theorem in Computer Science?

A: The fundamental principle of the Sampling Theorem is that a signal can be perfectly reconstructed from its samples if the sampling frequency is more than twice the highest frequency component of the signal.



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The Sampling Theorem states that a continuous-time signal can be perfectly reconstructed from its samples if the sampling frequency is above a certain finite limit. This is rarely achieved in practice. Lastly, the theorem inherently assumes infinite samples which is not realistic in real-world applications.

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What is the core principle of the Sampling Theorem in Computer Science?
The Sampling Theorem, also known as the Nyquist-Shannon theorem, provides a fundamental rule for converting continuous-time signals into discrete-time signals. It states that a signal can be perfectly reconstructed if the sampling frequency is more than twice the highest frequency component of the signal.
What is Quantisation and how does it relate to the Sampling Theorem?

What does the [Nyquist Shannon Sampling Theorem](#) guide in information technology and telecommunications?

The [Nyquist Shannon Sampling Theorem](#) guides the digital capturing of continuous signals without any loss of information, given that the sampling frequency is more than twice the signal's highest frequency.

What are the two fundamental concepts the [Nyquist Shannon Sampling Theorem](#) is built upon?

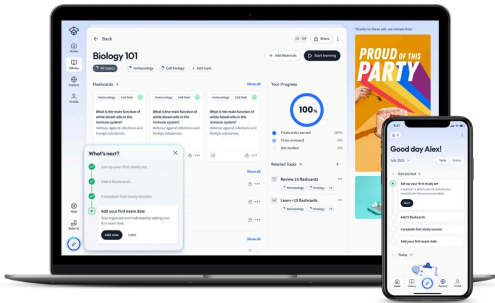
The [Nyquist Shannon Sampling Theorem](#) is fundamentally built upon two concepts: sampling and aliasing.

What does the [proof of the Sampling Theorem](#) use and what idea it establishes?

The [proof of the Sampling Theorem](#) or the [Nyquist-Shannon Theorem](#) uses principles of Fourier Transform and Euler's formula. It establishes the idea that an original signal can be recovered from its samples, when sampled above twice the maximum frequency of the original signal.

What are some practical implications of understanding the [proof of the Sampling Theorem](#)?

[Understanding the Sampling Theorem](#) enables effective data compression, the design of anti-aliasing filters to eliminate frequencies above maximum, replication of signals in telecommunication and broadcasting, and better use of devices like MRI scanners in medical imaging.



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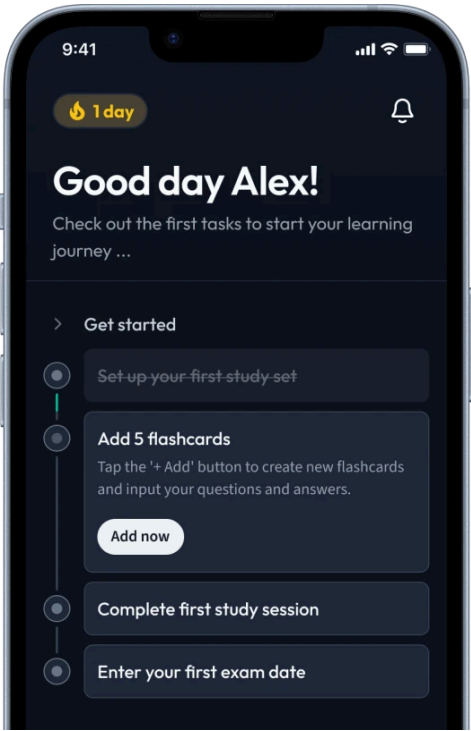
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